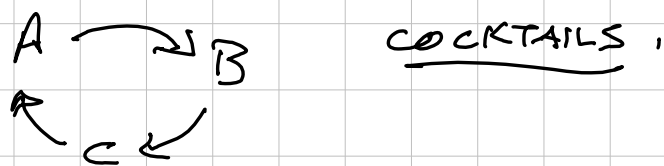


antibiotic resistance - MGMT.

- AGRICULTURAL use (limit)
 - medical use: limit unnecessary use.
- cycling? limiting?



- prob (multiple resistance alleles) << prob (indiv resistance)
- varying in SPACE.
- treating for longer. ~ does reduce chance of persistence
- → increases probability of COLLATERAL selection

(selecting non-pathogenic (commensal) bacteria for A/B resistance)

Tuberculosis.

VIRUSES.

- biochemistry is MORE similar to host biochem.
- Vaccination ~ priming immune system

RESISTANCE evolves by immune evasion rather than detoxification/pathway change

≈ distinction between RQ dynamics (recognition) and resistance/tolerance (effectors)

- STRAIN REPLACEMENT

wild type, alpha, delta, omega...

RECOMBINATION does happen...

H1N1, H3N2, ...

antigenic SHIFT/ DRIFT

HIV, single drug resistance evolves quickly.

mutation rate x turnover time

every single point mutation can occur within a few days.

- DRUGS. target "non-host-like" biochem.
- reverse transcriptase inhibitors
- integrase inhibitors

- HAART. • cocktail of chemicals with three different modes. (minimize side effects)

REDUCE virus load. keeps people healthy reduces prob of resistance evolution

